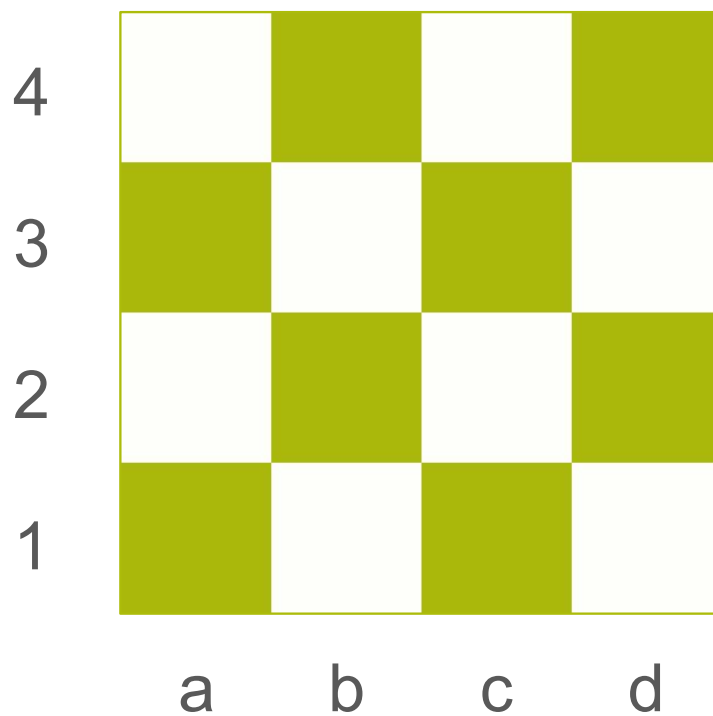


Base workbook

This document should be printed
2 pages-per-sheet

4 Queens

Can you fit 4 Queens on an 4x4 chessboard?



Write down the coordinates of each queen in your solution!



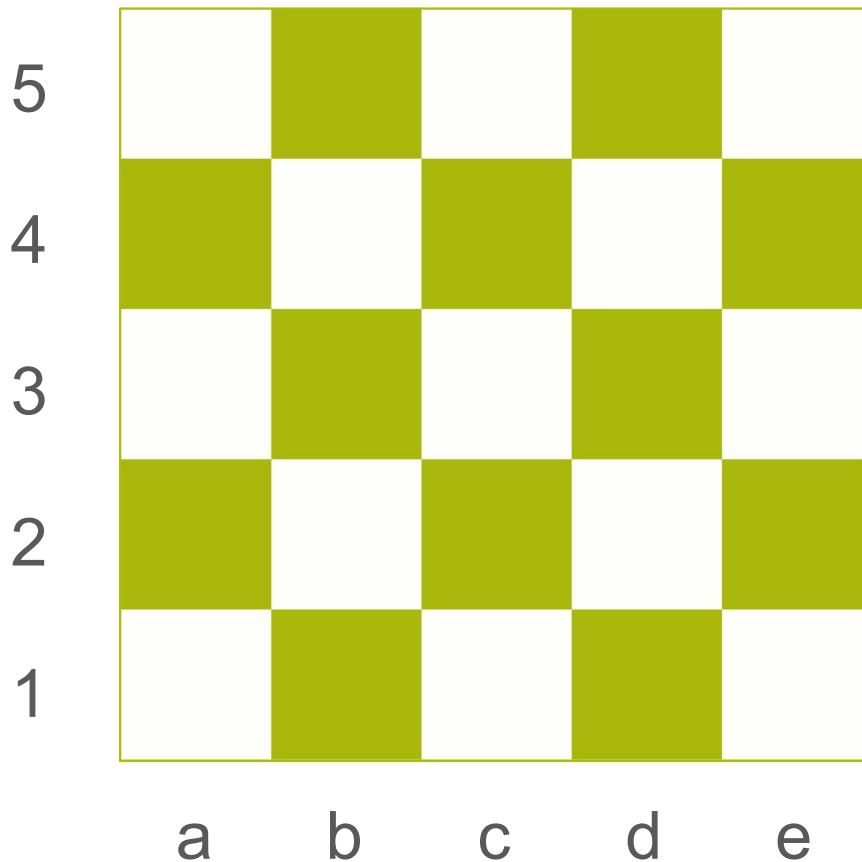






5 Queens

Can you fit 5 Queens on an 5x5 chessboard?



Write down the coordinates of each queen in your solution!





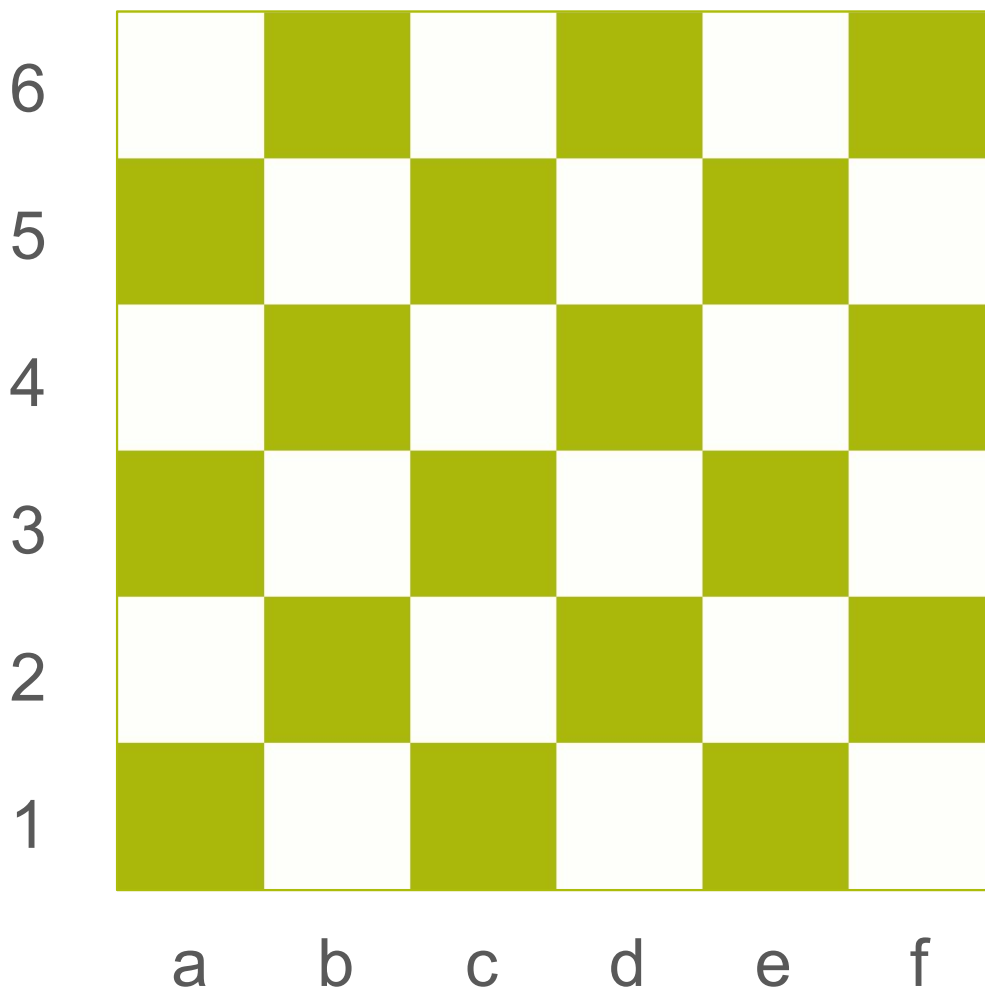






6 Queens

Can you fit 6 Queens on an 6x6 chessboard?



Write down the coordinates of each queen in your solution!







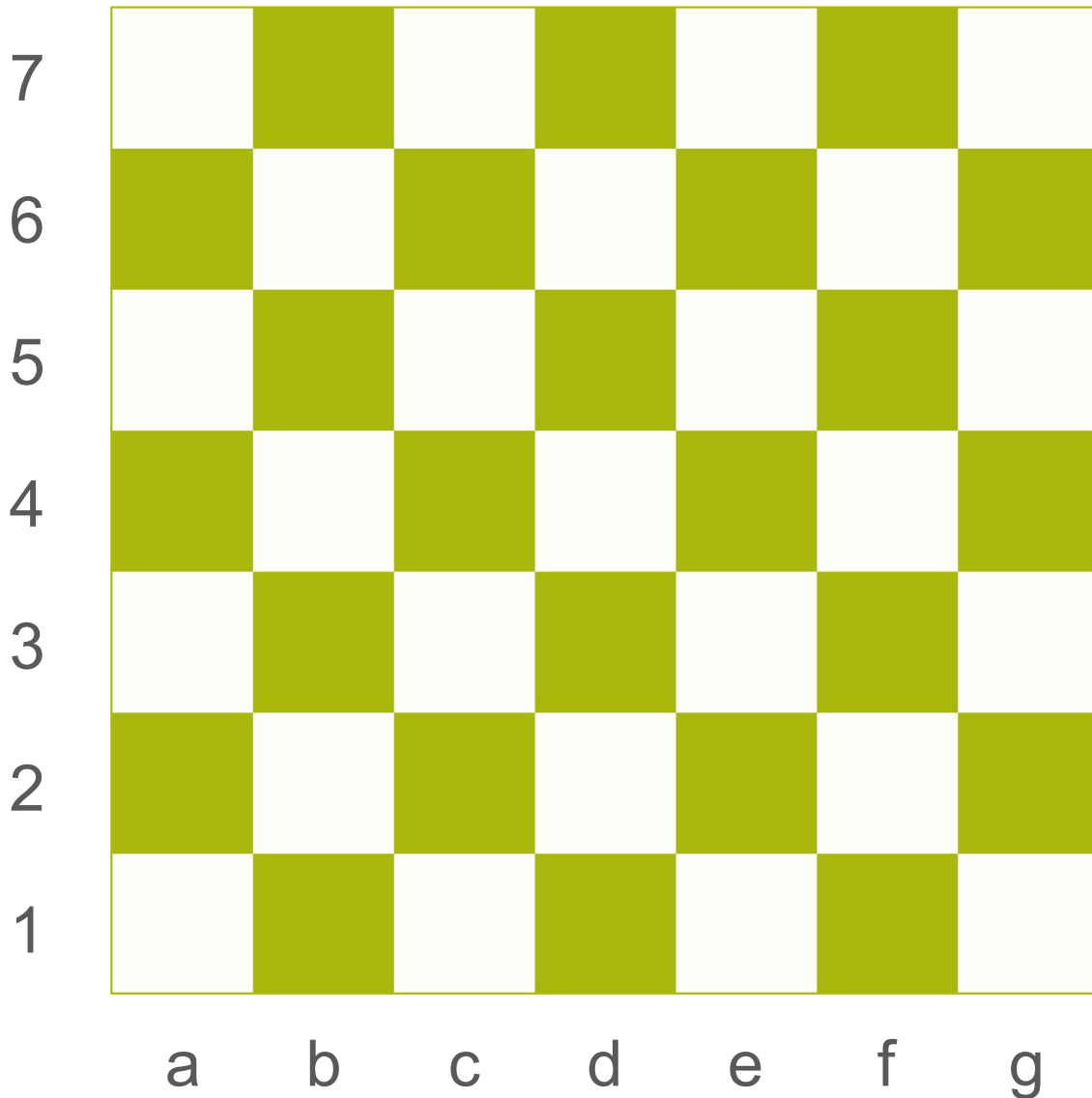






7 Queens

Can you fit 7 Queens on an 7x7 chessboard?



Write down the coordinates of each queen in your solution!









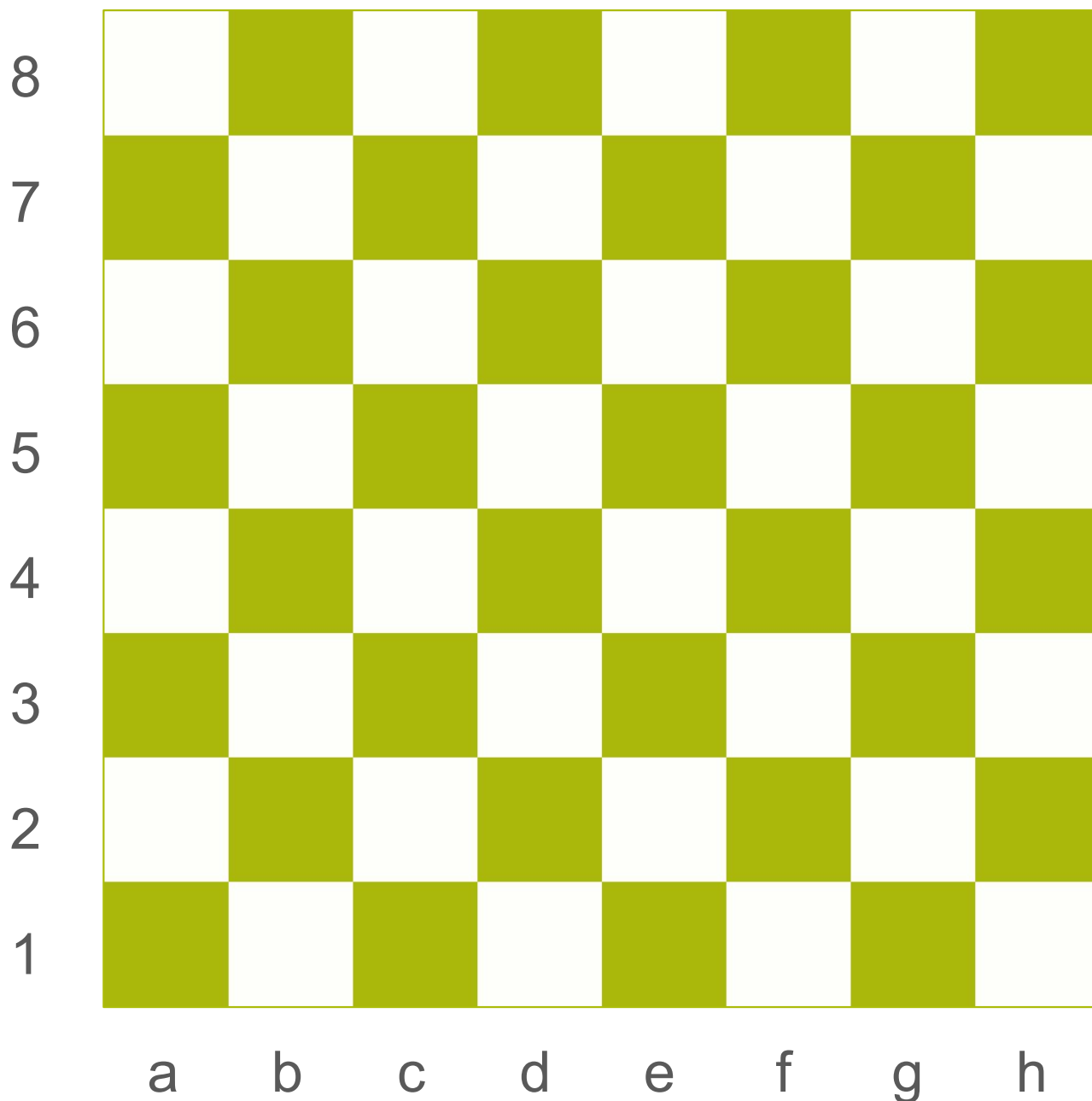






8 Queens

Can you fit 8 Queens on an 8x8 chessboard?



Write down the coordinates of each queen in your solution!

 _____  _____  _____  _____

 _____  _____  _____  _____

Bonus: Other Chess Pieces?

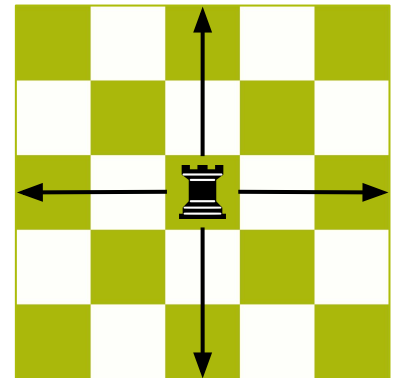
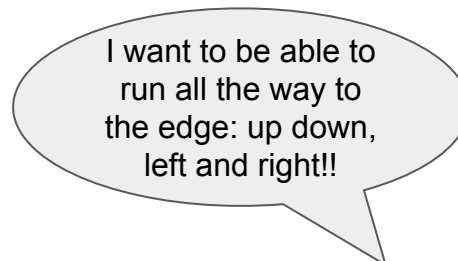
We know we can fit 8 queens on an 8x8 chessboard, but what happens if we change the chess piece?

Use the chessboard on the next page answer each proposed piece swap!

What if its a **Rook**?

How many fit?

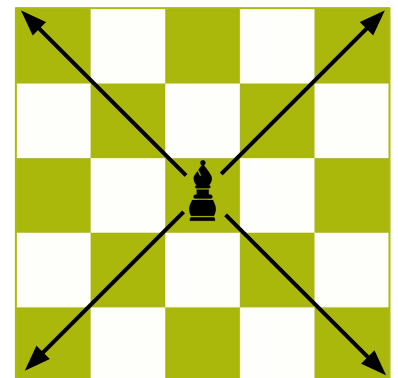
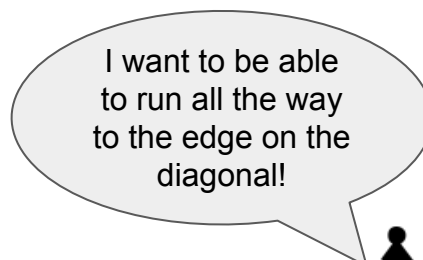
What are their coordinates?



What if its a **Bishop**?

How many fit?

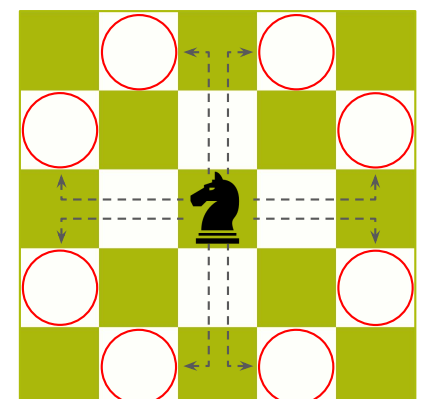
What are their coordinates?



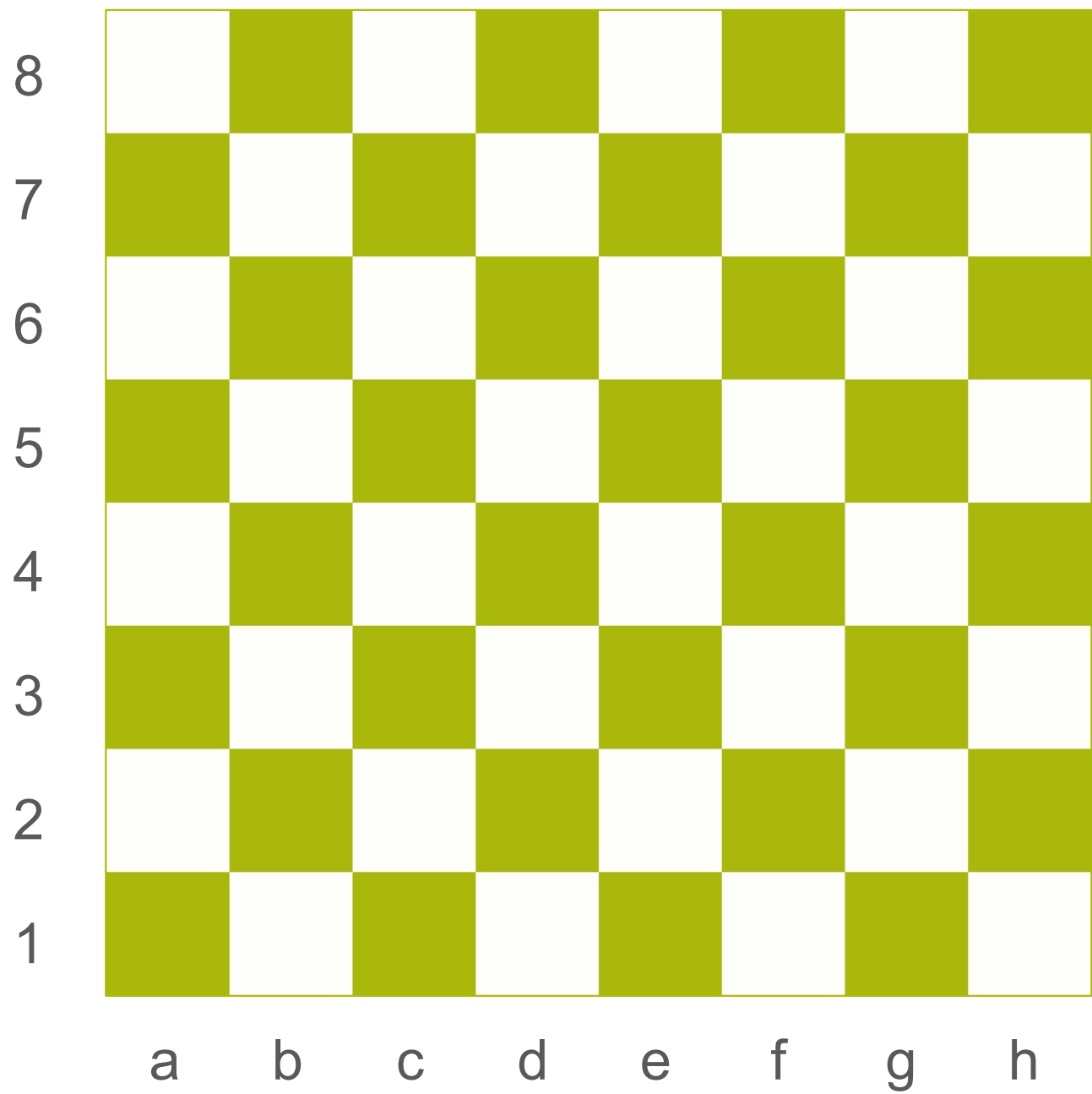
What if its a **Knight**?

How many fit?

What are their coordinates?



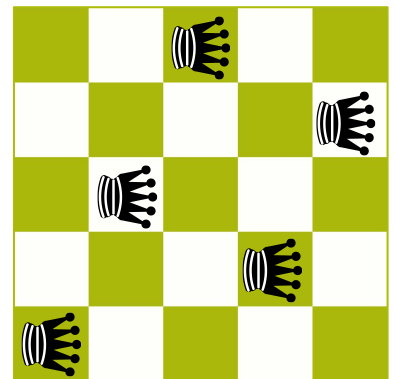
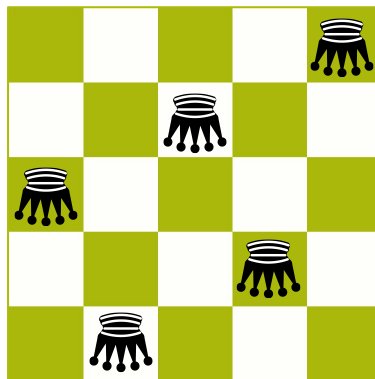
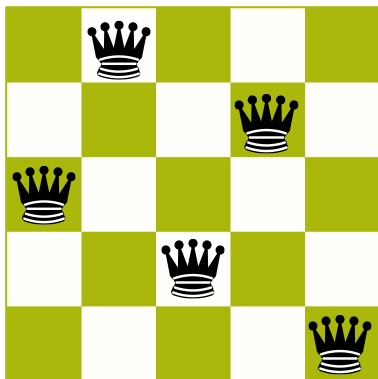
Bonus 8x8 Chessboard



Bonus: Multiple Solutions!

You may have noticed that for many of the problems you've completed there are *multiple* solutions

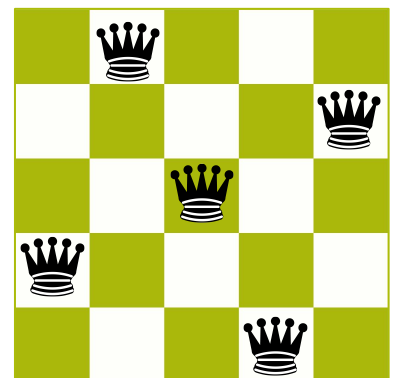
Look here's an example!



But wait, on first look they do look like *unique solutions*
However, we can see they are just variations!

2 is 1 flipped vertically!
3 is 1 rotated 90*!

This means they are isomorphic and are not *really* new solutions!
A new solution following would be one like this, which cannot be transformed to match the first, making it non-isomorphic!



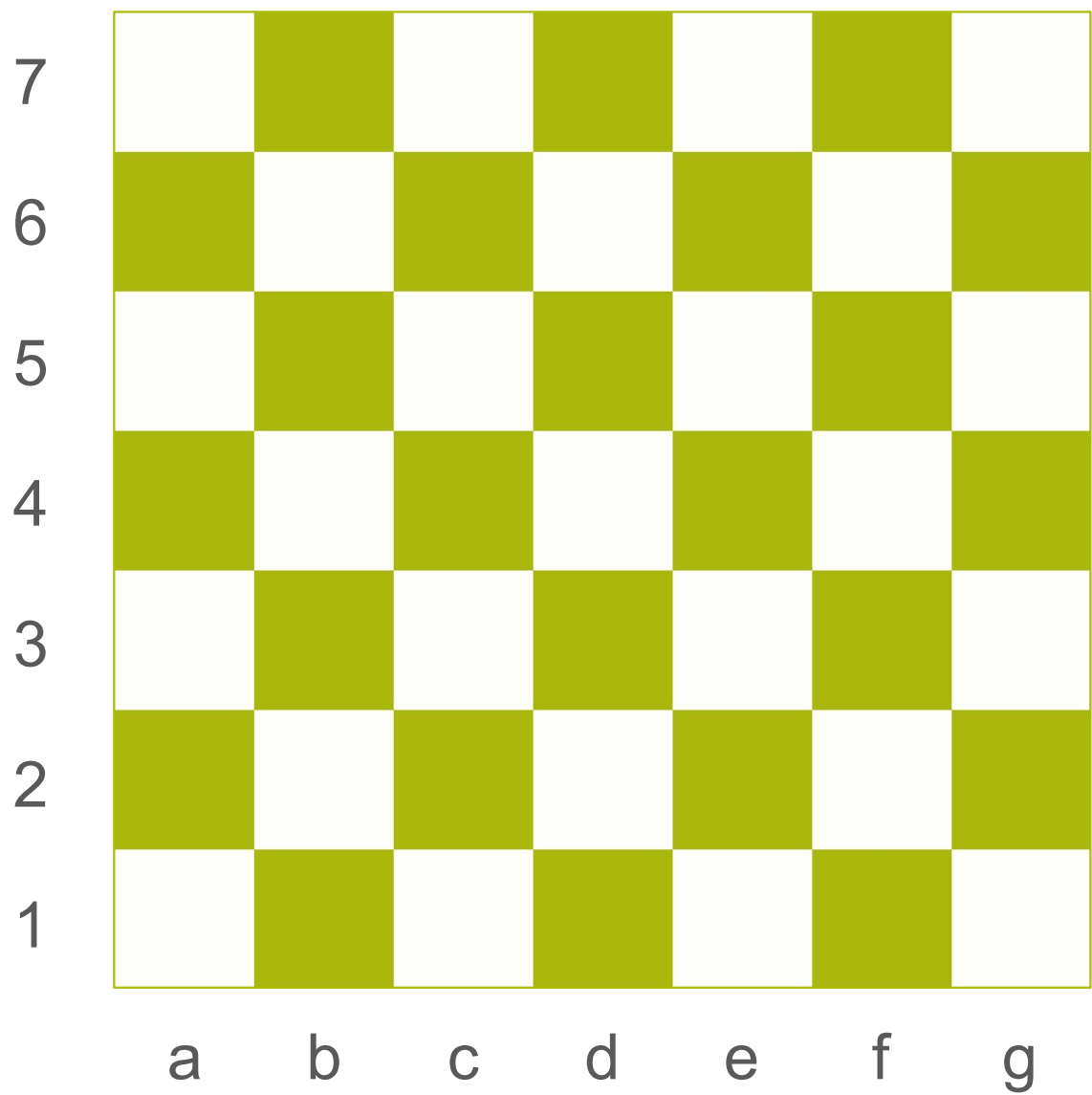
A new challenge:

Can you find more non-isomorphic solutions for the 8 queens problem?

Write out and get them checked by the tutors!

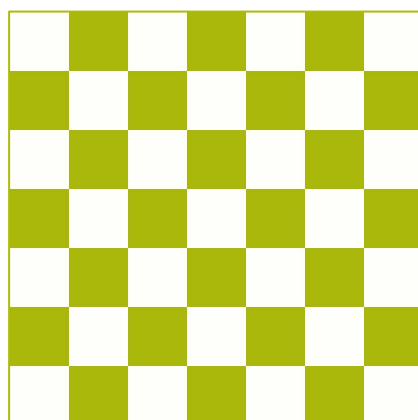
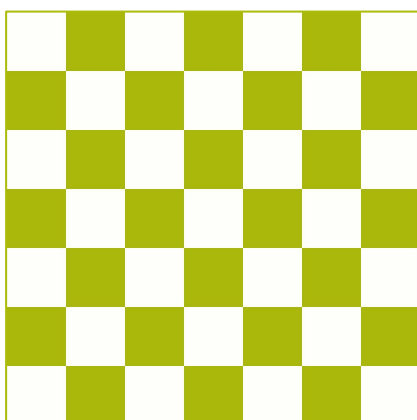
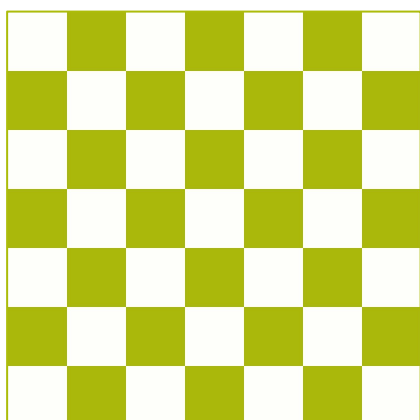
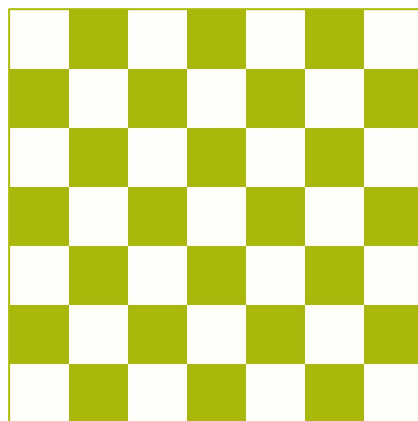
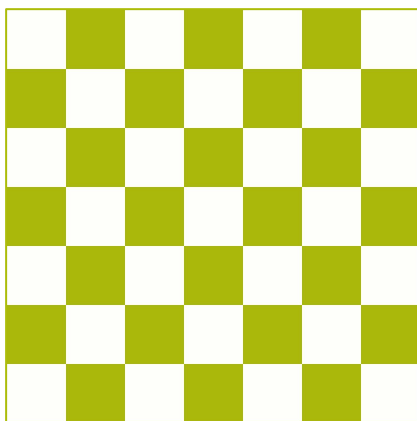
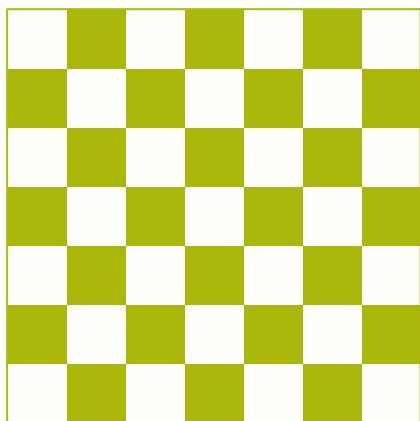
There is a big 8x8 board for you to find your solutions on, and a sheet with a few small chess boards for you to draw out your non-isomorphic solutions on!

Bonus 7x7 Chessboard

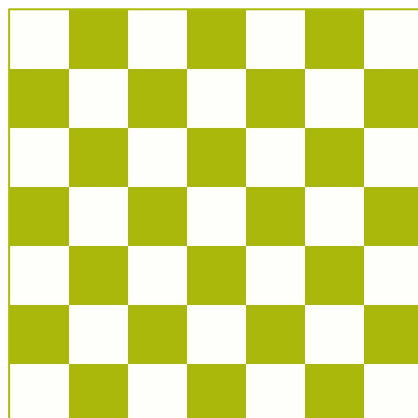
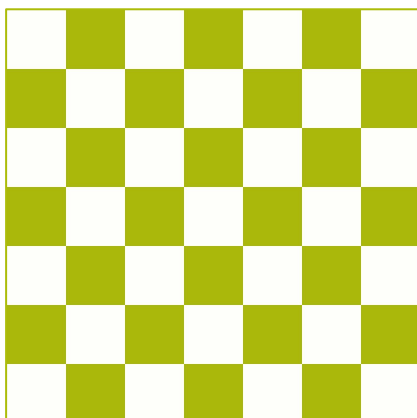
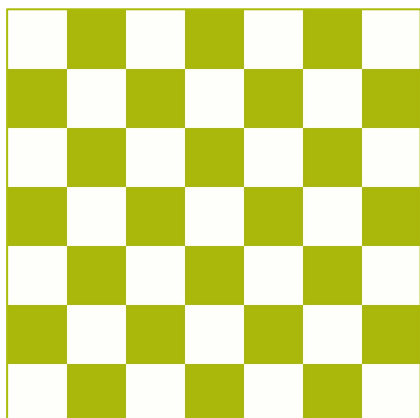


Bonus: Multiple Solutions; 7 Queens

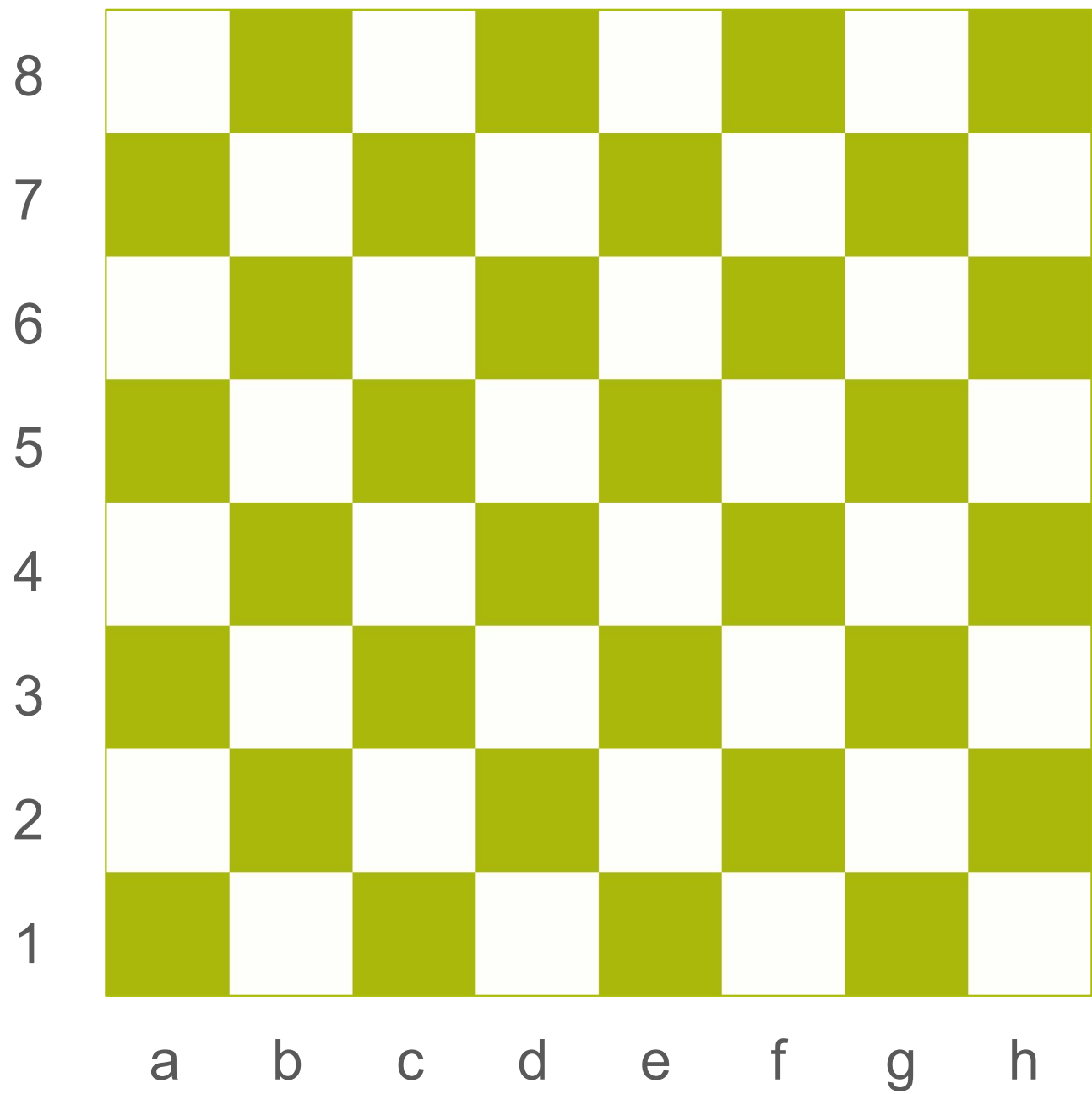
Fill in these boards as you find your unique solutions so you can check for isomorphism!



There are 6 unique solutions for 7 queens but here are some extra boards just in case!



Bonus 8x8 Chessboard



Bonus: Multiple Solutions; 8 Queens

Fill in these boards as you find your unique solutions so you can check for isomorphism! There are 12 total 8x8 unique solutions!

